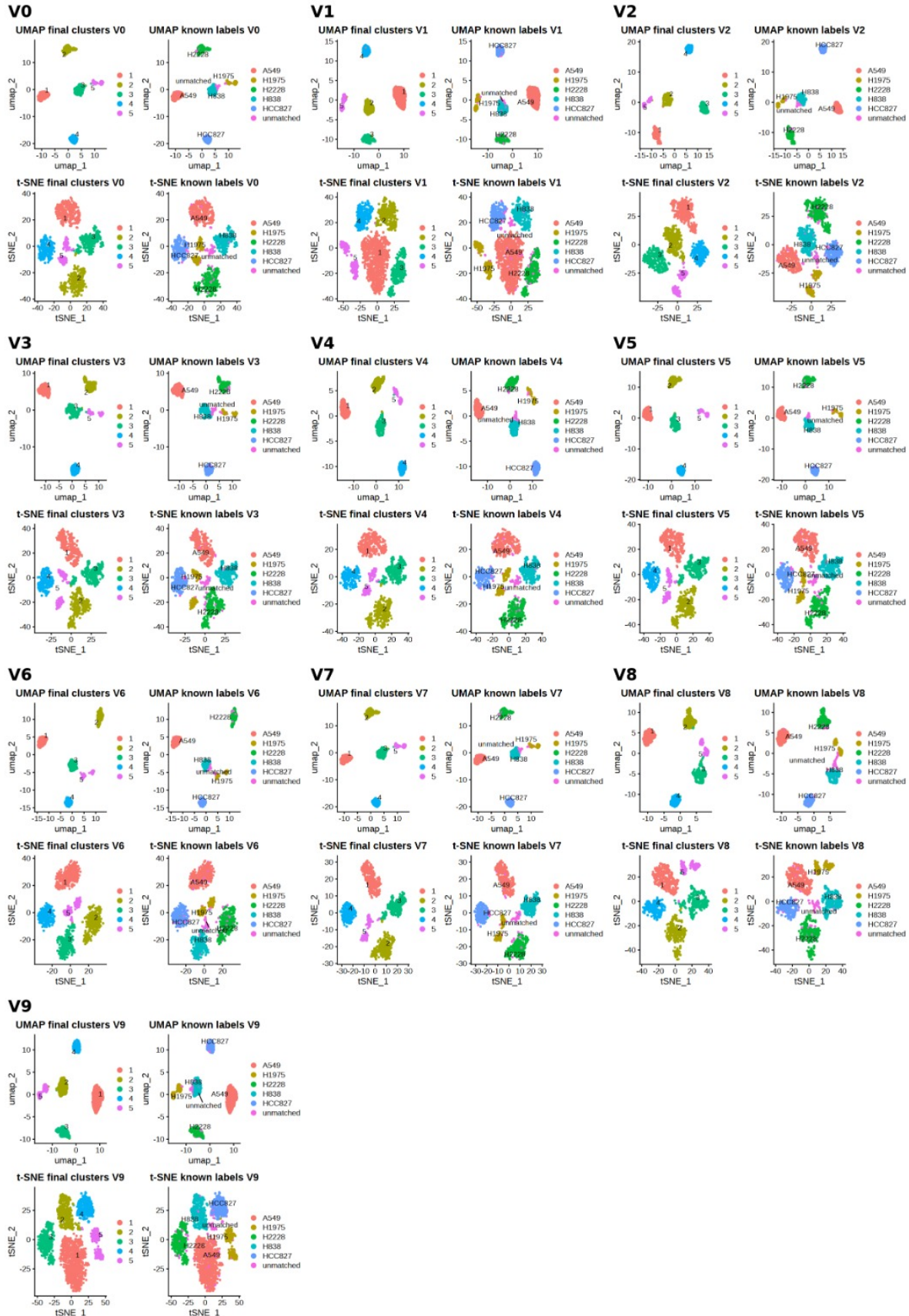


Supplementary material

1. scMixology supplementary figures

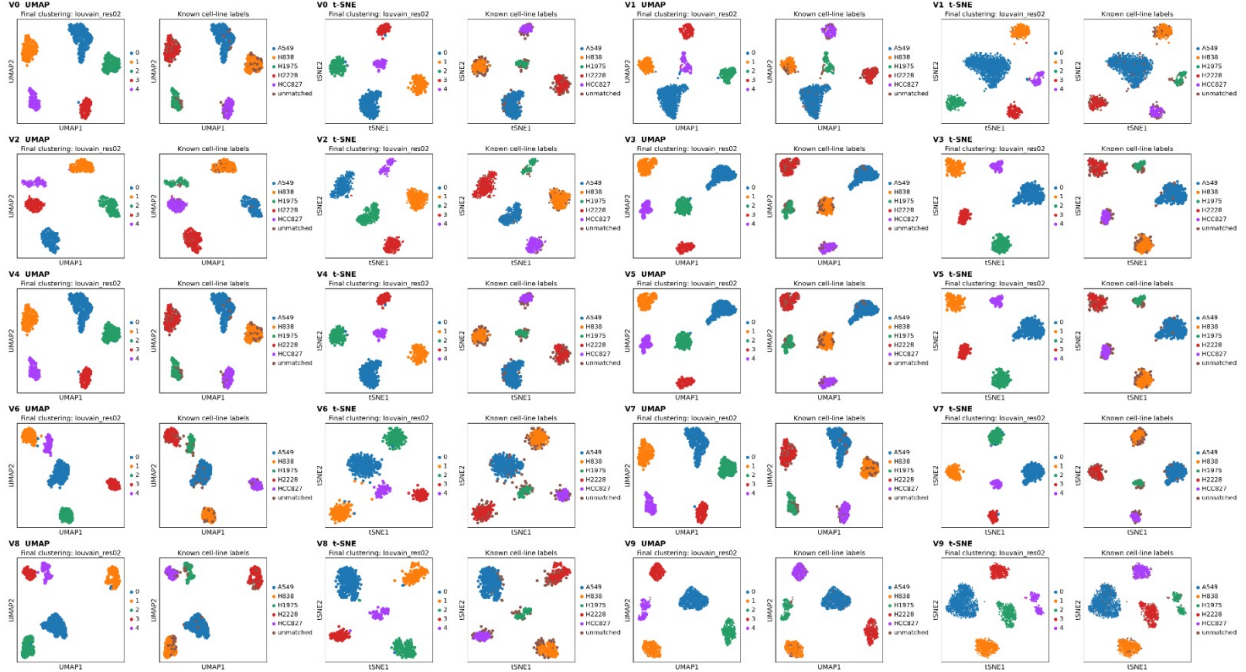
Supplementary Figure S1: UMAP and t-SNE panels for Seurat V0-V9

Seurat: UMAP and t-SNE embeddings, V0-V9

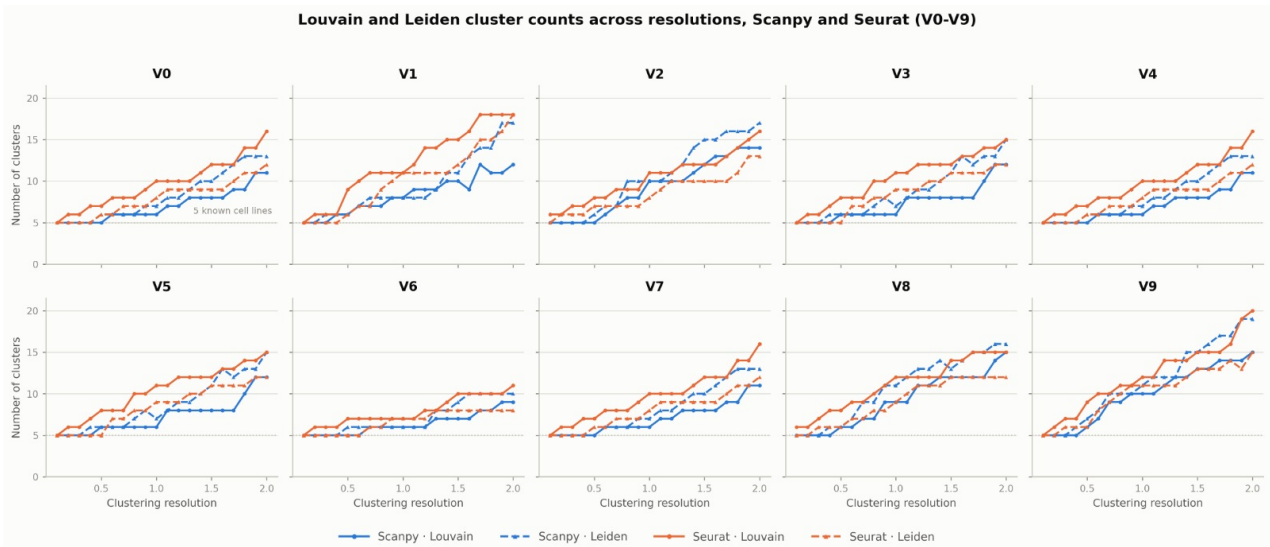


Supplementary Figure S2: UMAP and t-SNE panels for Scanpy V0-V9

Scanpy: UMAP and t-SNE embeddings, V0-V9

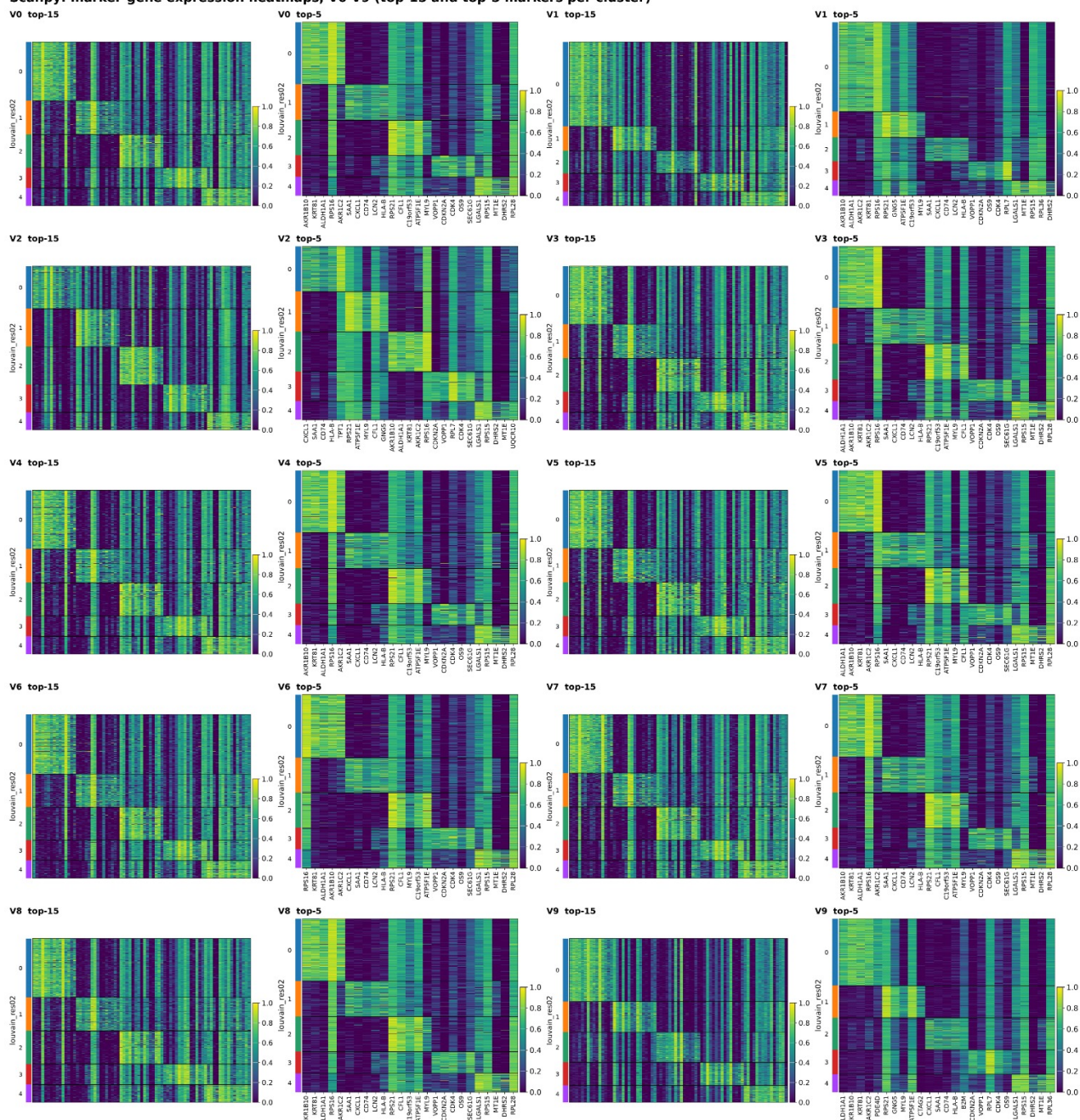


Supplementary Figure S3: Louvain and Leiden clusters counts across resolution, Scanpy and Seurat



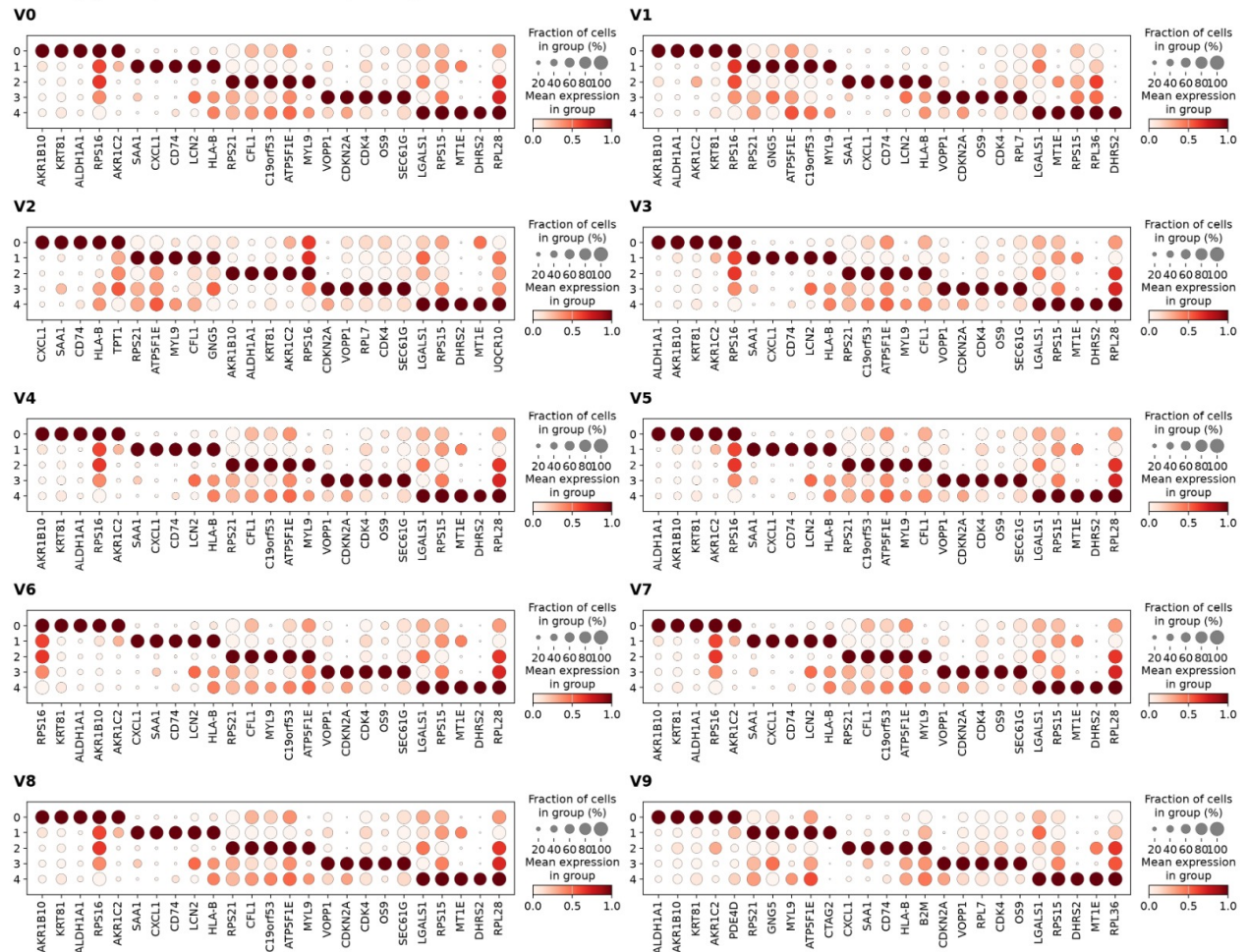
Supplementary Figure S4: Expanded scMixology marker heatmaps, top 15 and top5, Scanpy

Scanpy: marker-gene expression heatmaps, V0-V9 (top-15 and top-5 markers per cluster)



Supplementary Figure S5: Expanded scMixology top5 dotplot, Scanpy

Scanpy: top-5 marker dotplots, V0-V9



2 scMixology supplementary tables

- Supplementary Table S1: Overall comparison of the scMixology baseline and parameter-sensitivity versions.

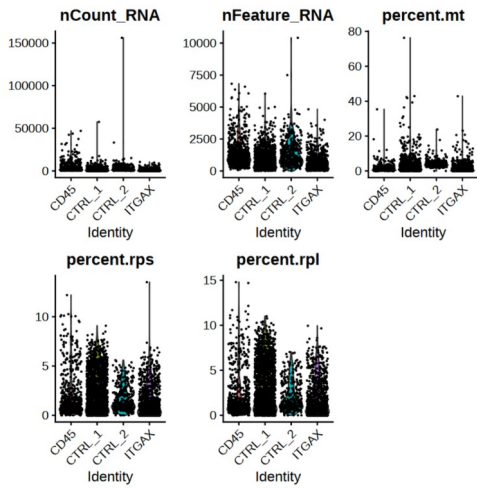
Version	Parameter setting	Change in analysis orientation	Purpose of the change	Verified result and status
V0	Baseline. Scanpy: 200-5,000 genes, mt<5%, Scrublet, 1,000 HVGs, 50 PCs, k=10, t-SNE perplexity 30, Leiden r=0.2. Seurat: 200-6,200 features, mt<5%, counts<60,000, 1,000 HVGs, 50 PCs, k=20, t-SNE perplexity 18, Leiden r=0.2.	Reference workflow from which all sensitivity versions were derived.	Establish a reproducible benchmark and recover the five known cell lines before changing individual decisions.	Scanpy: 774 singlets, 672 labelled, 5 clusters, ARI 1.0000. Seurat: 1,639 retained, 1,533 labelled, 5 clusters, ARI 0.9974.
V1	Only change: mitochondrial limit 5%→15%.	Relaxes cell-level QC and retains cells with higher mitochondrial RNA.	Test whether the 5% limit removes usable cells and whether added lower-quality cells weaken clustering.	Scanpy: 1,443 labelled, 5 clusters, ARI 0.9992. Seurat: 2,871 labelled, 5 clusters, ARI 0.9970.
V2	Only change: maximum genes/features→6,500.	Relaxes the upper complexity filter and retains high-feature cells, including possible doublets.	Measure the retention-versus-purity effect of the upper feature threshold.	Scanpy: 1,751 labelled, 5 clusters, ARI 0.9976. Seurat: 1,759 labelled, 6 clusters, ARI 0.9823 at r=0.2; 5 clusters and ARI 0.9991 at r=0.08.
V3	Only change: compute and use 30 PCs.	Reduces the dimensions used to build the neighbor graph and clusters.	Test whether major cell-line structure is captured with fewer components and less noise.	Scanpy: 672 labelled, 5 clusters, ARI 1.0000. Seurat: 1,533 labelled, 5 clusters, ARI 0.9974.
V4	Intended: compute 100 PCs and use 50. Scanpy file used 100 downstream; Seurat used 100/50.	Expands the representation; actual Scanpy setting also changes the neighbor graph to 100 PCs.	Separate the effect of computing extra PCs from using them downstream. The submitted Scanpy notebook cannot make that separation.	Scanpy: 672 labelled, 5 clusters, ARI 1.0000; operationally duplicates V6. Seurat: 1,533 labelled, 5 clusters, ARI 0.9974.
V5	Only change: compute 50 PCs and use 30.	Keeps the full PCA calculation while restricting neighbor graph, UMAP and t-SNE inputs.	Test whether removing lower-ranked PCs improves robustness or reduces unnecessary variation.	Scanpy: 672 labelled, 5 clusters, ARI 1.0000. Seurat: 1,533 labelled, 5 clusters, ARI 0.9974.
V6	Only change: compute and use 100 PCs.	Adds lower-ranked PCs to graph construction and downstream embeddings.	Test whether additional dimensions add signal or introduce noise and over-fragmentation.	Scanpy: 672 labelled, 5 clusters, ARI 1.0000. Seurat: 1,533 labelled, 5 clusters, ARI 0.9974.
V7	Only change: t-SNE uses 30 PCs and perplexity 40; graph unchanged.	Changes the t-SNE visualization but leaves the precomputed neighbor graph and clusters unchanged.	Demonstrate that visualization settings can alter appearance without changing clustering evidence.	Scanpy: 672 labelled, 5 clusters, ARI 1.0000. Seurat: 1,533 labelled, 5 clusters, ARI 0.9974; clustering unchanged because only t-SNE changed.
V8	Only change: increase HVGs from 1,000 to 2,000.	Expands the feature space before scaling and PCA.	Test whether additional variable genes improve separation or add noise and computation.	Scanpy: 672 labelled, 5 clusters, ARI 1.0000. Seurat: 1,533 labelled, 5 clusters, ARI 0.9989.
V9	Combined changes: mt<15% and maximum genes<6,500.	Simultaneously relaxes two cell-level QC filters and substantially expands the analyzed population.	Test a permissive QC scenario; it is not a one-parameter experiment.	Scanpy: 3,142 labelled, 5 clusters, ARI 0.9996. Seurat: 3,165 labelled, 5 clusters, ARI 0.9970.

3 mIEC supplementary figures

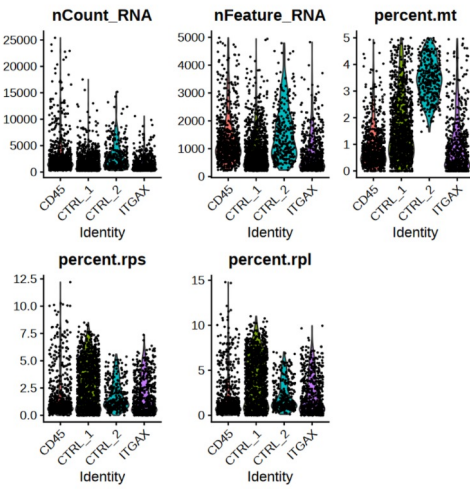
- Supplementary Figure S6: mIEC - QC distributions before and after filtering for all four workflows

Seurat | No integration

A Before filtering | n = 3,494

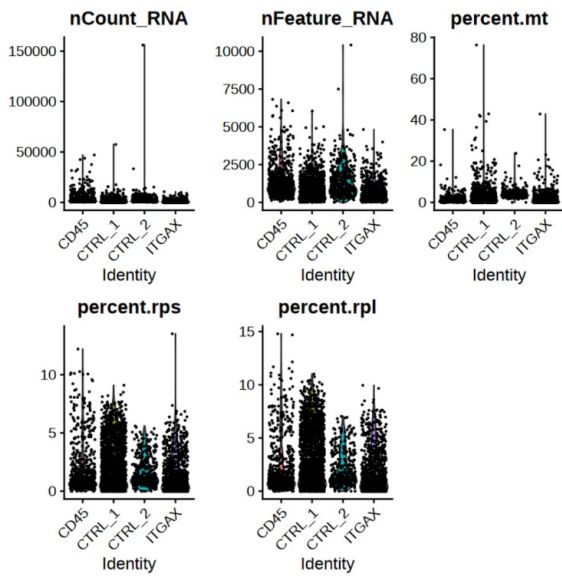


B After QC filtering | n = 2,741

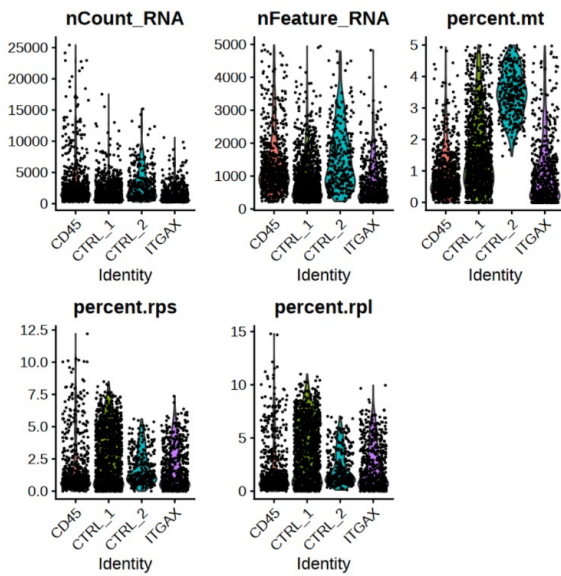


Seurat | CCA

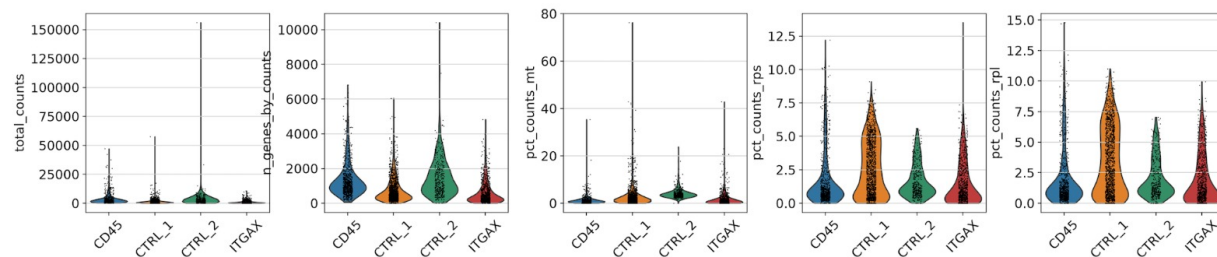
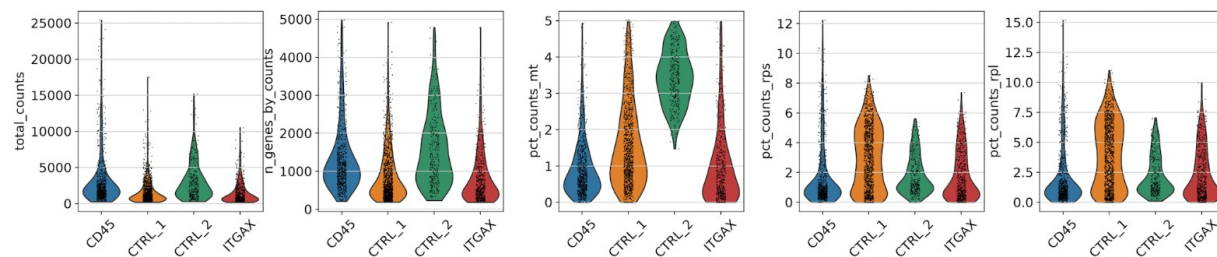
A Before filtering | n = 3,494



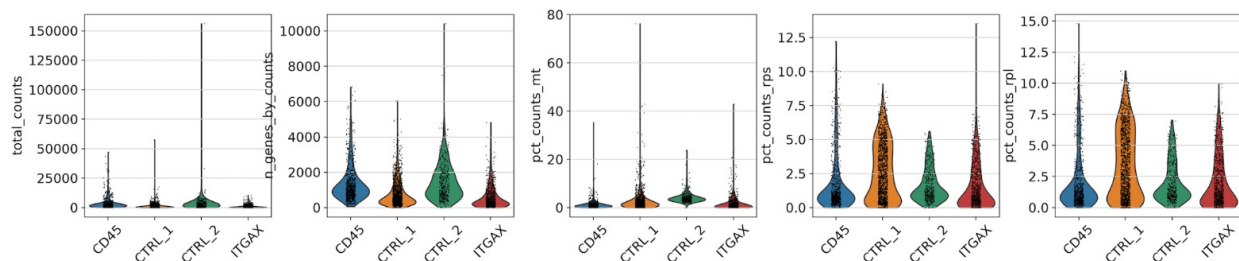
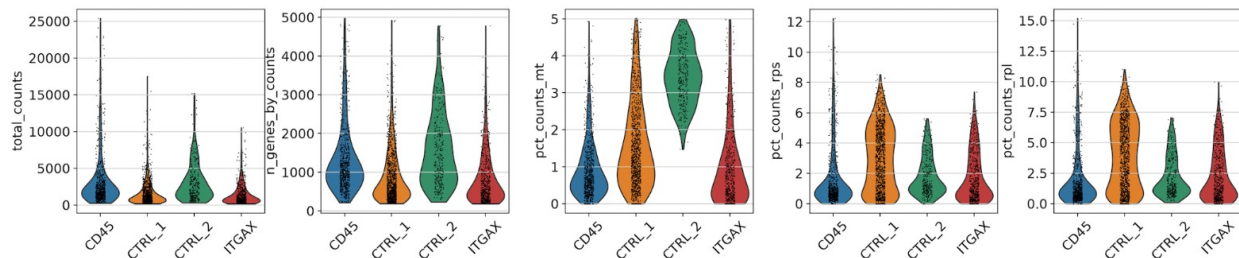
B After QC filtering | n = 2,741



Five metrics: total counts, detected genes, mitochondrial %, RPS %, and RPL %, grouped by sample. Before doublet removal; native y-axis scales retained.

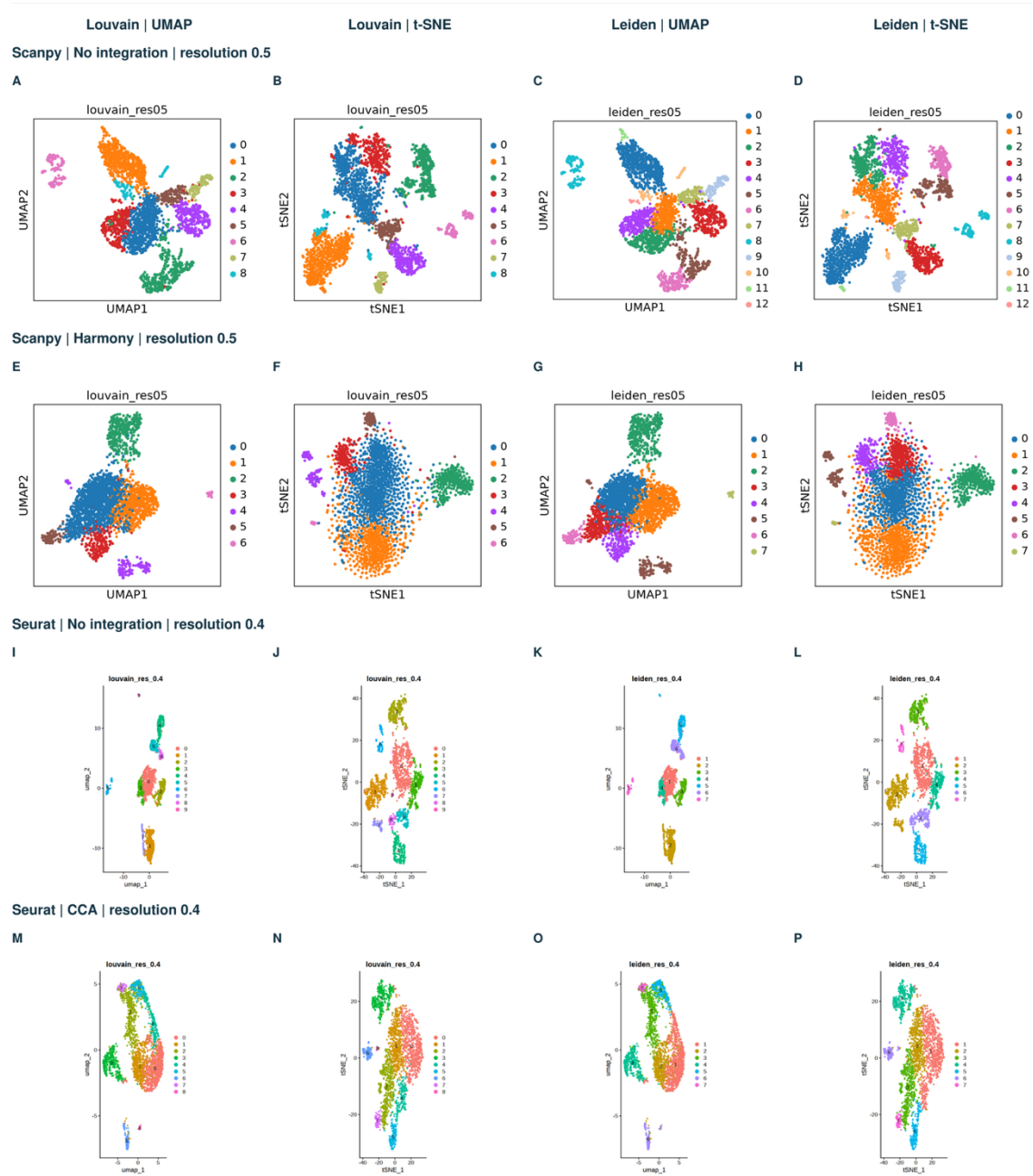
A Before filtering | n = 3,496**B After QC filtering | n = 2,741**

Five metrics: total counts, detected genes, mitochondrial %, RPS %, and RPL %, grouped by sample. Before doublet removal; native y-axis scales retained.

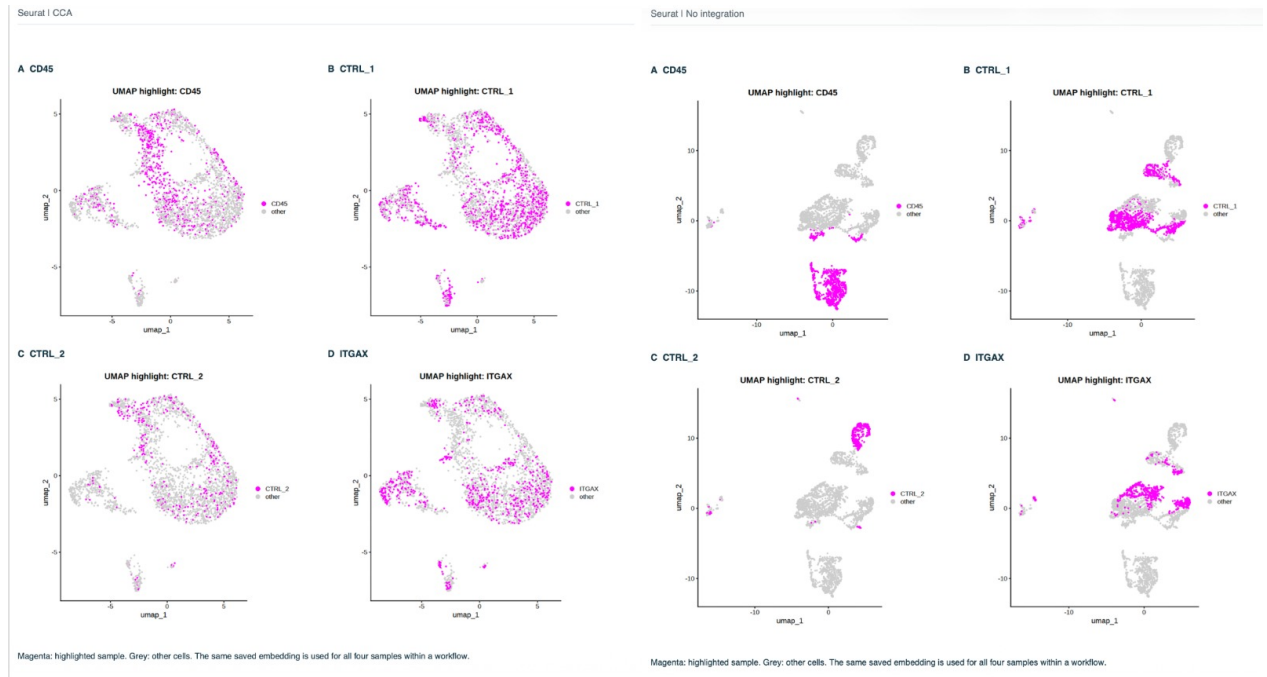
A Before filtering | n = 3,496**B After QC filtering | n = 2,741**

Five metrics: total counts, detected genes, mitochondrial %, RPS %, and RPL %, grouped by sample. Before doublet removal; native y-axis scales retained.

- Supplementary Figure S7: mIEC - one multi-panel Louvain/Leiden UMAP and t-SNE comparison for all workflows



- Supplementary Figure S8: mIEC - Seurat sample-highlight UMAPs



- Supplementary Figure S9: mIEC - Scanpy sample-highlight UMAPs



4 mIEC supplementary tables

• Supplementary Table S2 : mIEC - Complete shared-cell overlap matrices for Scanpy and Seurat
Scanpy: no integration → Harmony (n = 2,740 shared cells)

Source / target	0	1	2	3	4	5	6	Total
0	316	379	3	1	0	0	2	701
1	565	11	3	2	0	15	0	596
2	21	15	348	3	0	1	0	388
3	148	158	0	15	0	1	0	322
4	104	78	2	91	0	2	0	277
5	90	3	0	72	0	0	0	165
6	1	0	0	0	124	0	0	125
7	14	0	0	15	0	63	0	92
8	3	54	0	0	0	0	17	74
Total	1262	698	356	199	124	82	19	2740

Seurat: no integration → SCT-CCA (= 2,656 shared cells)

Source / target	0	1	2	3	4	5	6	7	8	Total
0	356	271	31	10	25	2	1	0	3	699
1	6	189	252	14	4	6	0	18	0	489
2	1	3	18	321	0	0	0	1	0	344
3	280	4	0	6	41	0	0	0	0	331
4	64	46	99	7	42	22	0	3	0	283
5	1	0	45	0	54	80	0	0	0	180
6	0	3	1	0	0	0	117	0	1	122
7	23	2	6	0	32	32	0	4	0	99
8	0	1	15	1	0	10	0	65	0	92
9	1	1	0	0	0	0	0	0	15	17
Total	732	520	467	359	198	152	118	91	19	2656

Rows are clusters without integration; columns are clusters after integration. Final clusters use Louvain resolution 0.5 in Scanpy and 0.4 in Seurat.

- Supplementary Table S3 : mIEC- Large comparison of the four mIEC analysis workflows.

Feature	Scanpy - no integration	Scanpy - Harmony	Seurat - no integration	Seurat - SCT-CCA
Data structure	AnnData; sparse counts, then dense scaled HVGs	Same AnnData structure plus corrected PC matrix	Seurat v5 assays/layers; scaled HVGs	RNA, SCT, and integrated assays
Doublet tool	Scrublet on combined matrix	Same Scrublet calls	Sample-aware scDblFinder	Same scDblFinder calls
Final cells	2,740	2,740	2,657	2,657
Normalization	10,000-count normalization + log1p	Same log-normalized expression	LogNormalize; scale factor 10,000	Per-sample SCTransform; regress mt%
Features	2,000 dispersion-based HVGs	Same 2,000 HVGs	2,000 VST HVGs	2,000 SCT integration features
Integration	None	Harmony by sample in 100-PC space	None	CCA anchors, dims 1-30; SCT; k.weight 70
Downstream PCs	20 uncorrected PCs	Harmony-corrected representation	20 uncorrected PCs	20 PCs of integrated assay
Graph/UMAP	10-neighbor Scanpy graph; umap-learn	10-neighbor Harmony graph	Seurat KNN/SNN; uwot/Annoy defaults	Seurat KNN/SNN on integrated PCA
Final clustering	Louvain r=0.5; 9 clusters	Louvain r=0.5; 7 clusters	Louvain r=0.4; 10 clusters	Louvain r=0.4; 7 clusters
Filtered marker records	5,376	4,685	8,389	4,346
Marker method	Wilcoxon on adata.raw; adjusted P<0.01	Same expression matrix; Harmony clusters	FindAllMarkers/Wilcoxon on RNA	FindAllMarkers/Wilcoxon on RNA
Main advantage	Transparent condition structure; flexible Python workflow	Fast low-dimensional correction; expression preserved	Cohesive, familiar R workflow; detailed assay handling	Strong multi-sample framework; SCT modeling and anchors
Main limitation	Merged Scrublet run may under-call; sample effects retained	Risk of overcorrection when sample=condition	Implicit defaults and different singlet set	Complexity; assumes shared states; sample-condition confounding
Best thesis role	Primary unintegrated sensitivity reference	Integrated Scanpy sensitivity analysis	Primary Seurat biological reference	Integrated Seurat sensitivity analysis